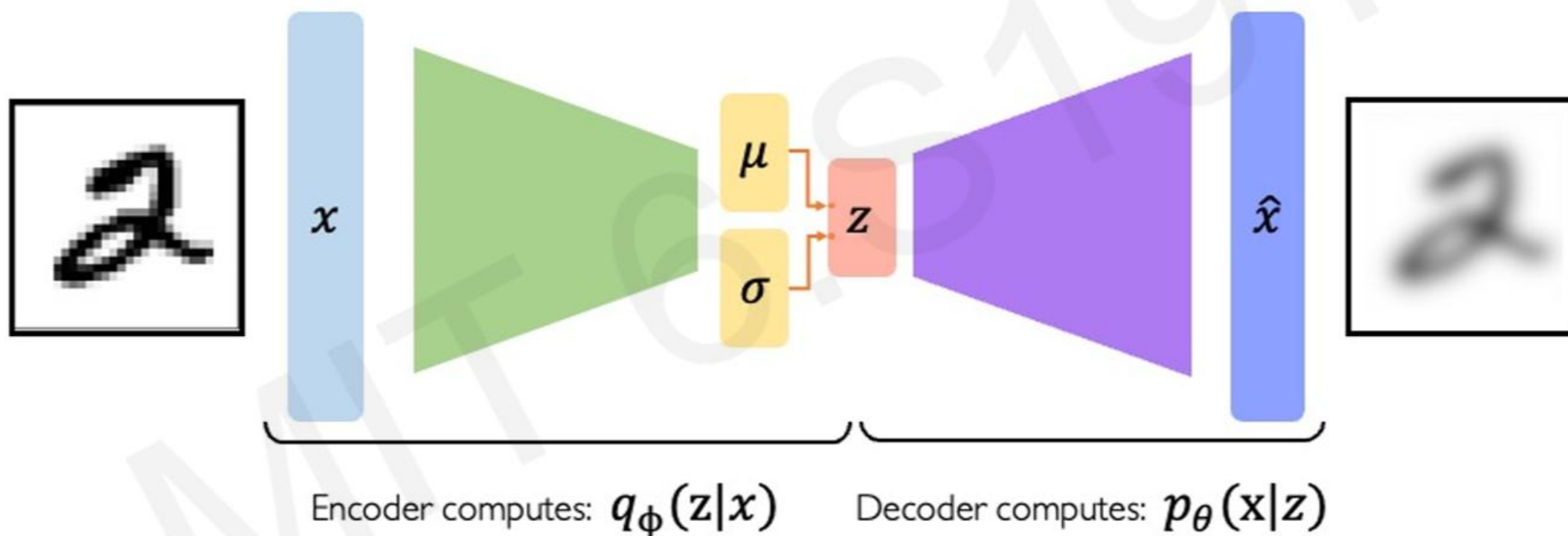
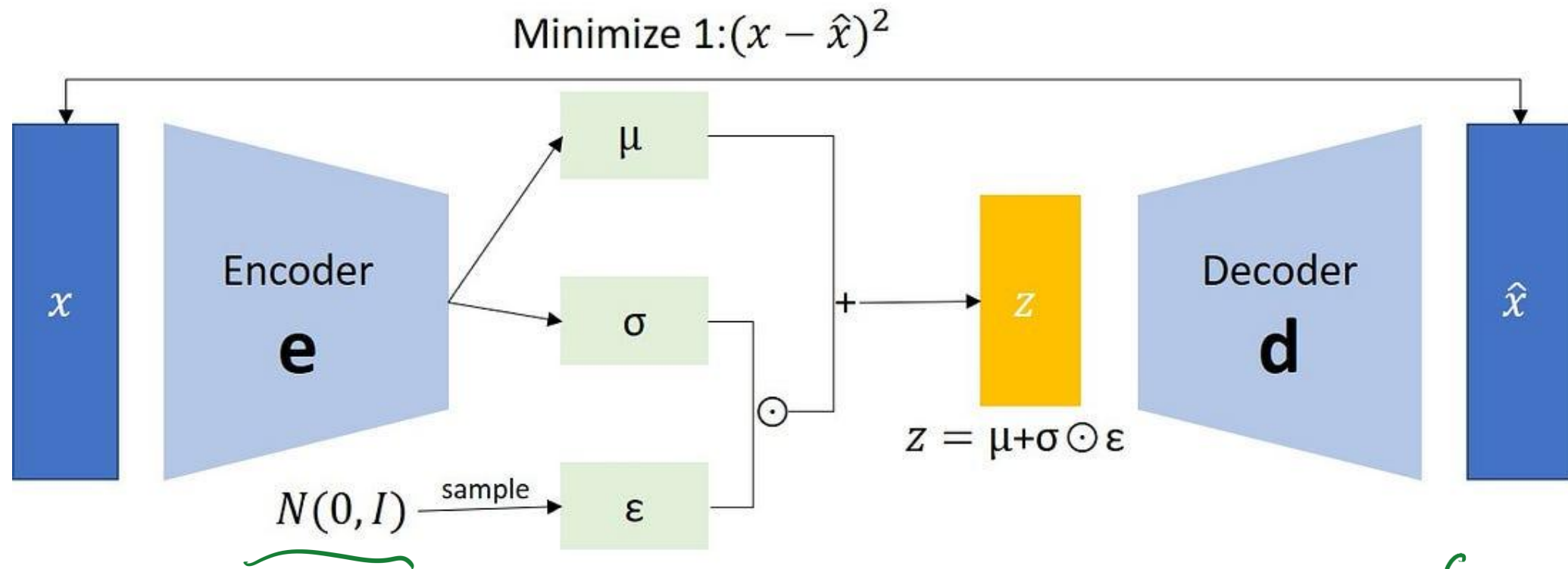


VAE optimization

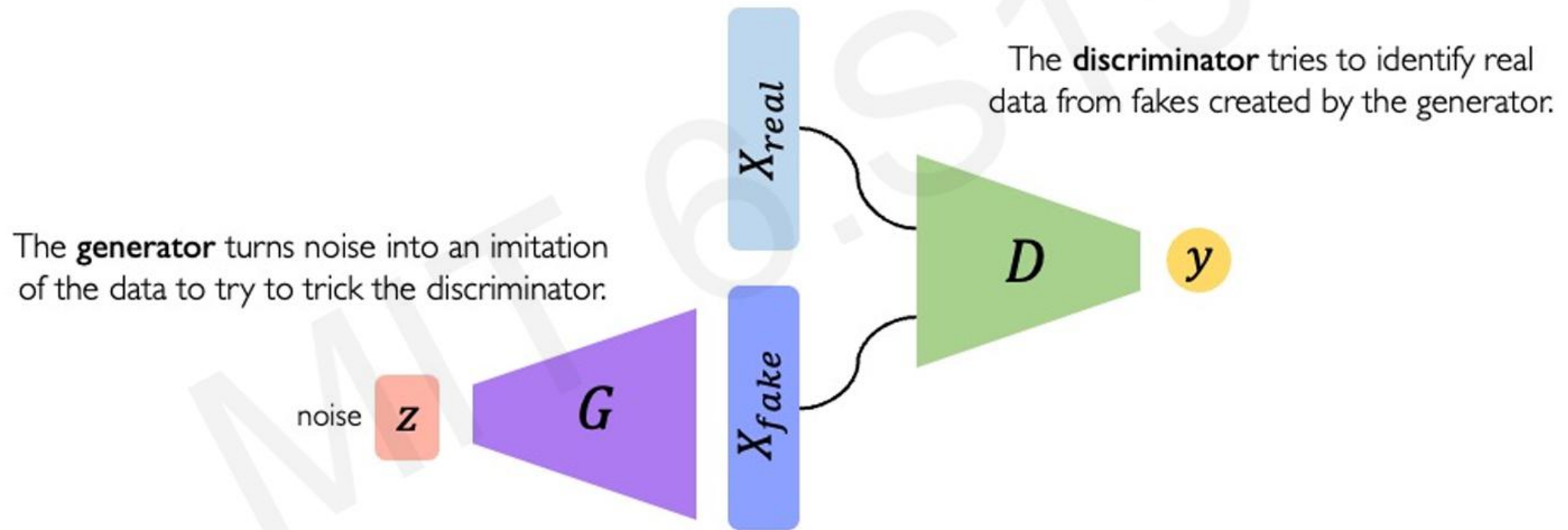
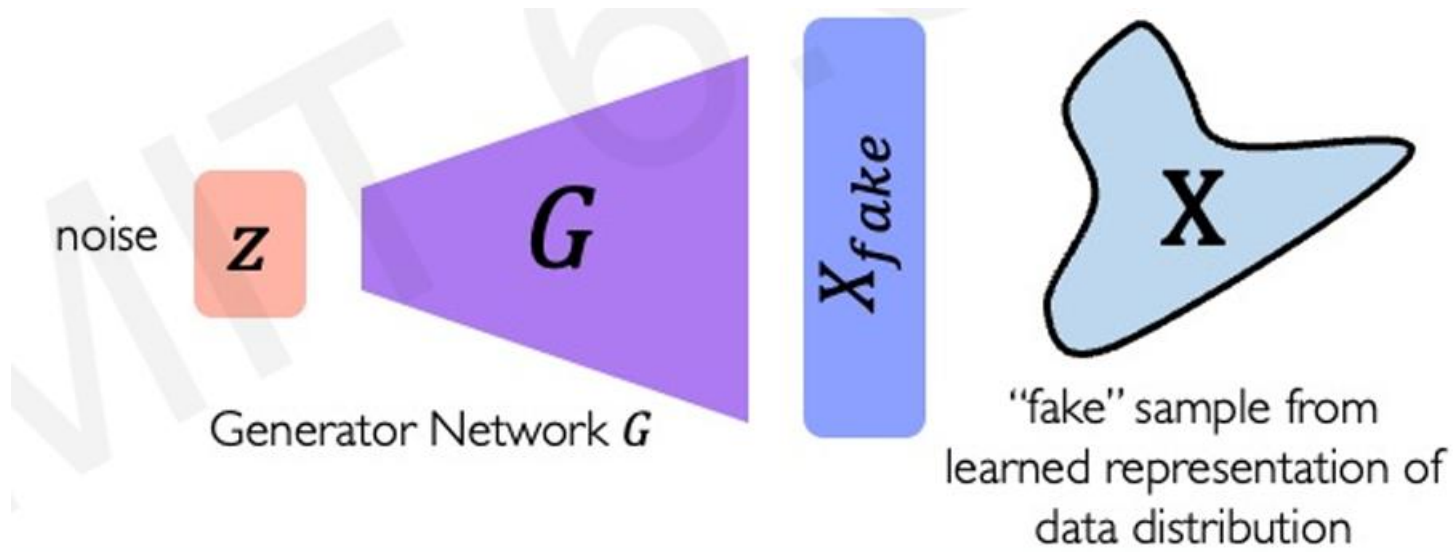




Minimize 2: $\frac{1}{2} \sum_{i=1}^N (\exp(\sigma_i) - (1 + \sigma_i) + \mu_i^2)$

$$z = \mu + \sigma \odot \varepsilon$$

$$\varepsilon = \frac{n - \mu}{\sigma}$$



Standard GAN loss function (min-max GAN loss)

$$E_x [\log(D(x))] + E_z [\log(1 - D(G(z)))]$$

The generator tries to minimize this function while the discriminator tries to maximize it

Discriminator loss

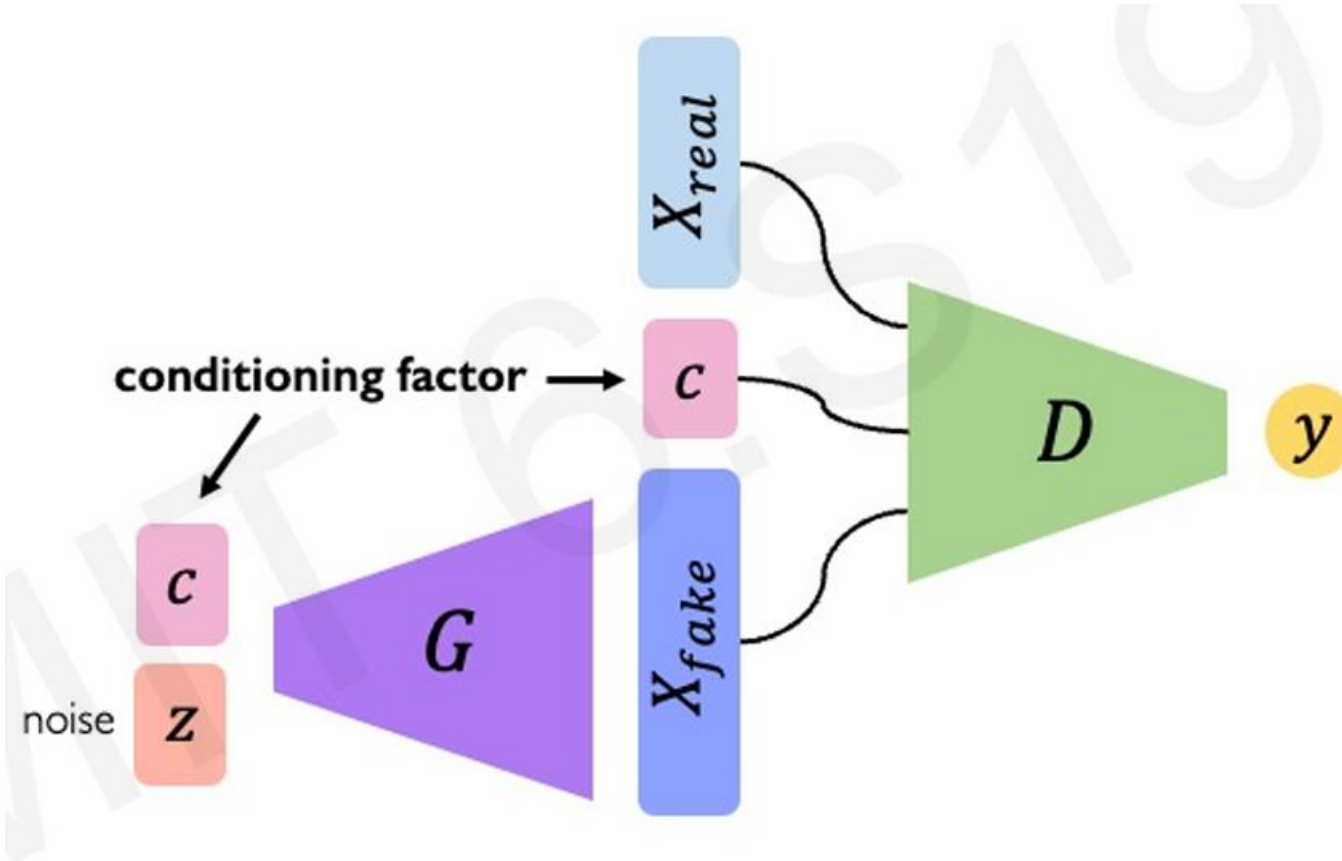
$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^m \left[\log D \left(\mathbf{x}^{(i)} \right) + \log \left(1 - D \left(G \left(\mathbf{z}^{(i)} \right) \right) \right) \right]$$

- **log(D(x))** refers to the probability that the generator is rightly classifying the real image,
- maximizing **log(1-D(G(z)))** would help it to correctly label the fake image that comes from the generator.

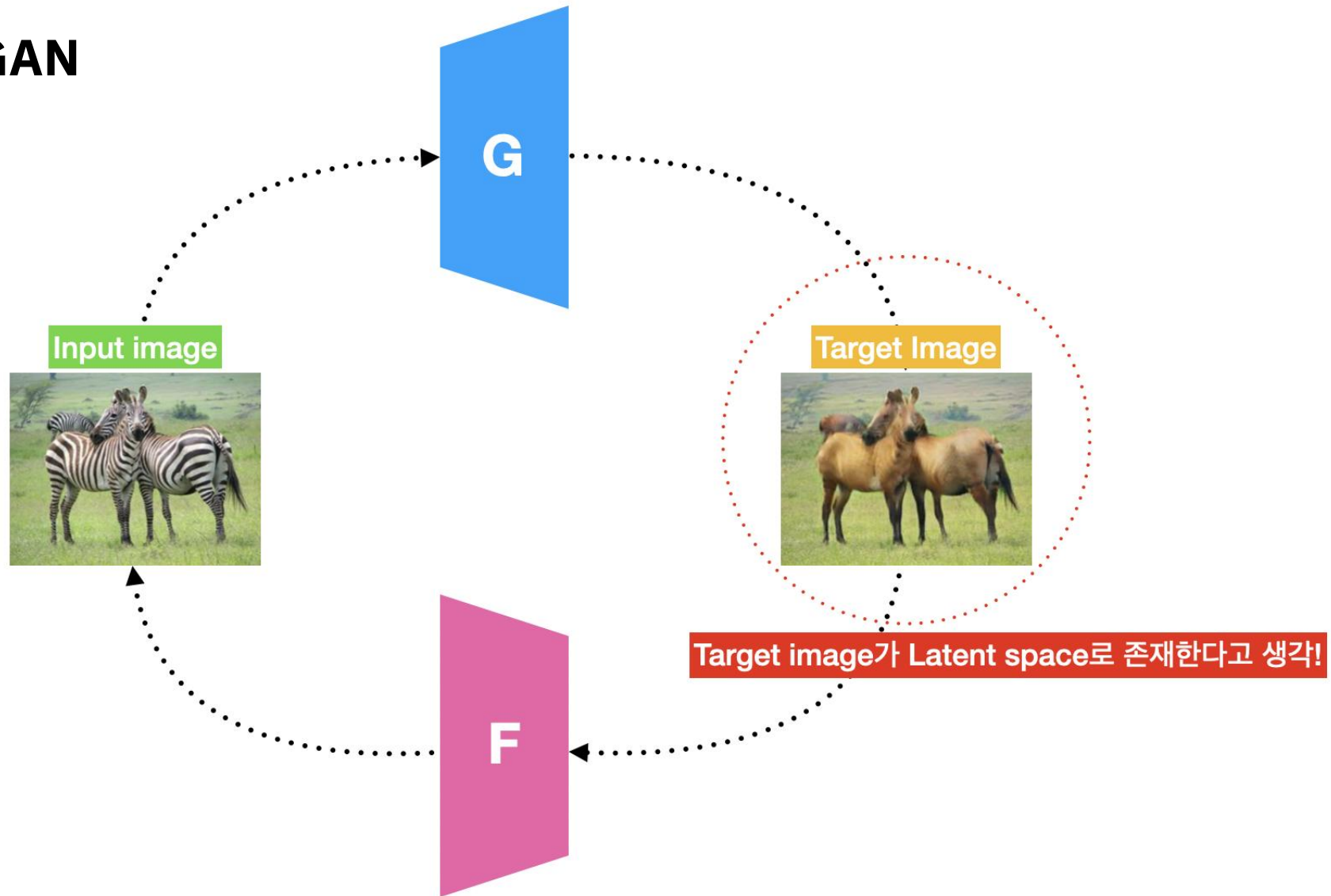
The following equation is minimized to training the generator:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^m \log \left(1 - D \left(G \left(\mathbf{z}^{(i)} \right) \right) \right)$$

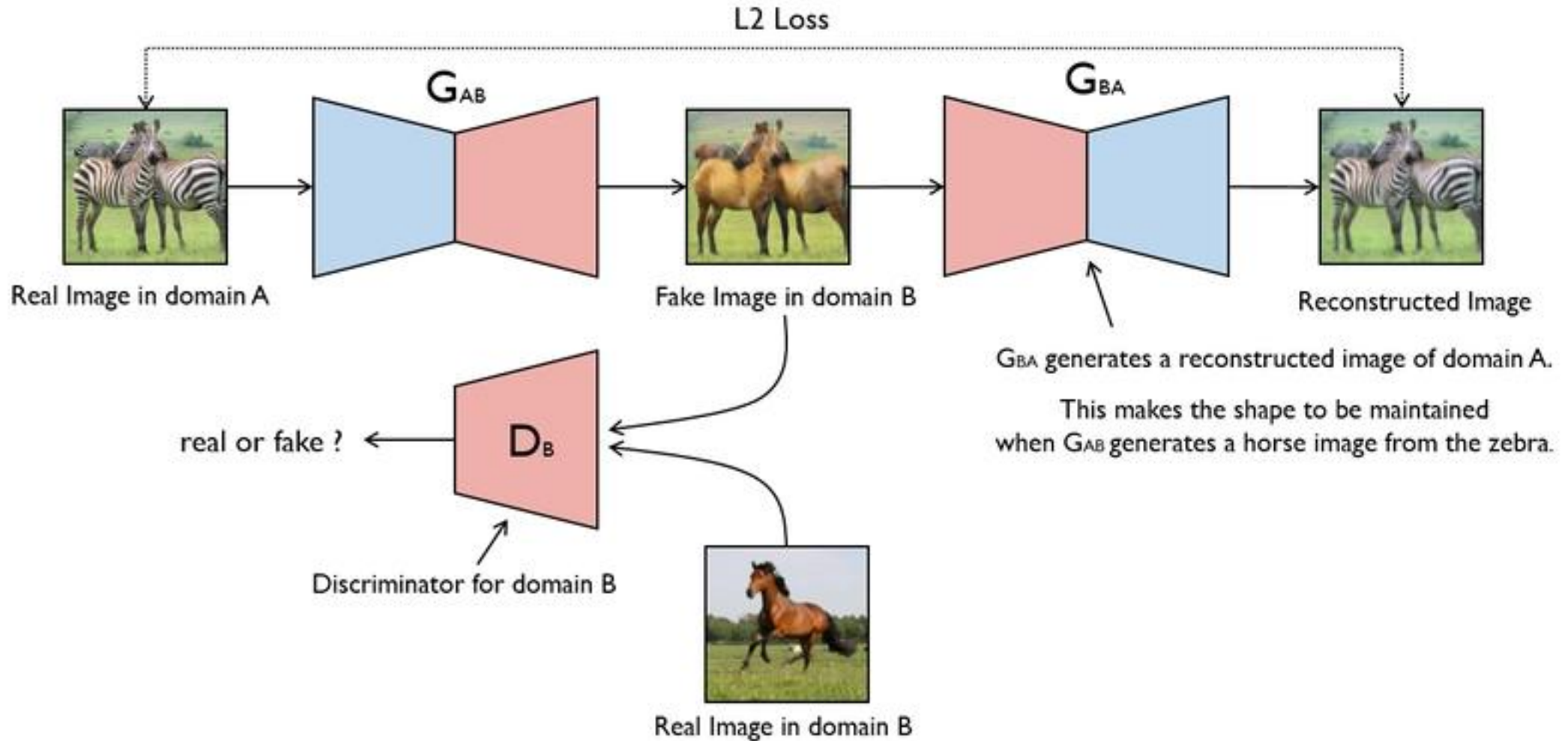
Conditional GAN



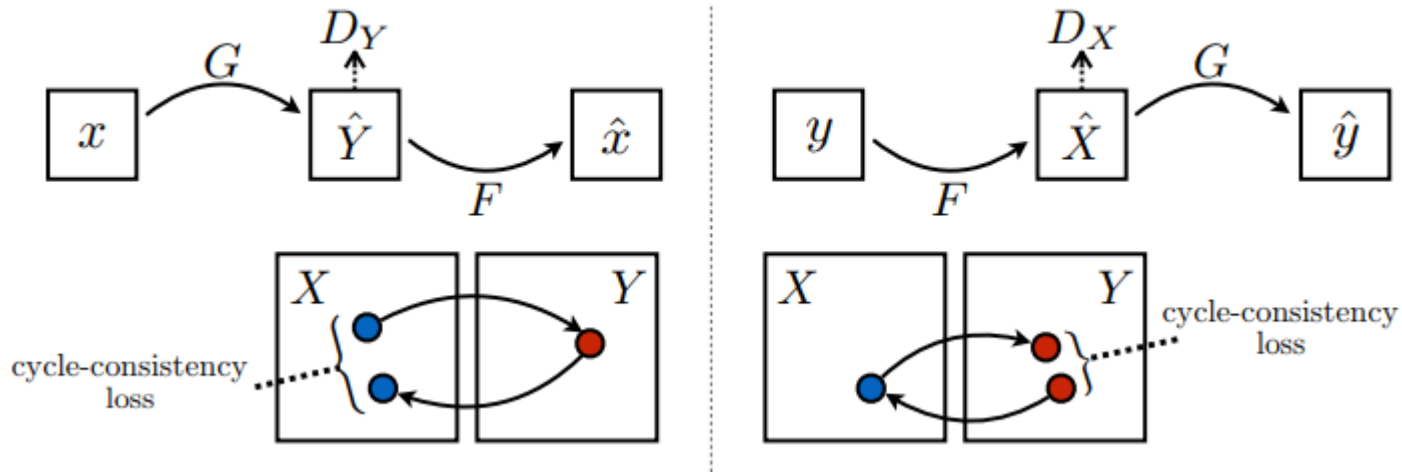
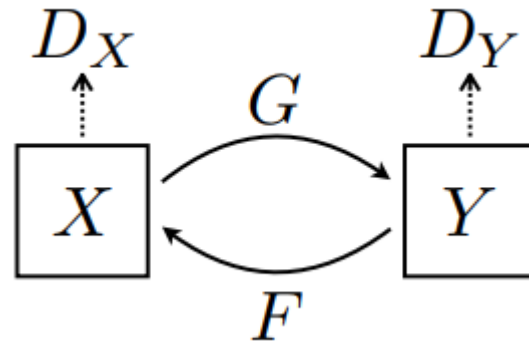
Cycle GAN



Cycle GAN



Cycle GAN loss



$$\text{Identity loss} = |G(Y) - Y| + |F(X) - X|$$

